1.4 $V_1 = 200\,km/h$, $S_1 = 1200\,m$, $S_2 = 50\,m$, $V_2 = 18\,km/h = 5\,m/s$

$V_1^2 = 2\cdot\frac{g}{2}x_1$ ①

$V_1 = \frac{g}{2}t_1$ ②

$S_1 - x_1 - S_2 = V_1 t_2$ ③

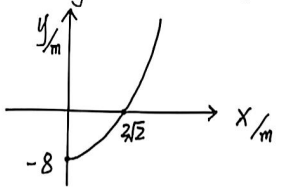
$S_2 = V_2 t_3$ ④

$t = t_1 + t_2 + t_3$ ⑤

解 $t = 36.36\,s$

此跳伞运动员在空中一共经历了 36.36s

1.7 (1) $\begin{cases} x = 2t \\ y = 4t^2 - 8 = (2t)^2 - 8 \end{cases}$ 可知 $y = x^2 - 8$ (m)



(2) 当 $t_1 = 1s$ 时，$x = 2m$，$y = -4m$ 故质点位置为 $(2m, -4m)$

$V_x = \dot{x} = 2\,m/s$

$V_y = \dot{y} = 8t = 8\,m/s$

故 $V = \sqrt{V_x^2 + V_y^2} = 2\sqrt{17}\,m/s$

方向与x轴夹角为θ 且 $\tan\theta = \frac{1}{4}$

$a_x = \dot{V}_x = 0\,m/s^2$

$a_y = \dot{V}_y = 8\,m/s^2$

故 $a = \sqrt{a_x^2 + a_y^2} = 8\,m/s^2$

方向沿y轴正方向

$\begin{cases} \vec{r}_1 = 2\vec{i} - 4\vec{j} \\ \vec{V}_1 = 2\vec{i} + 8\vec{j} \\ \vec{a}_1 = 8\vec{j} \end{cases}$

当 $t_2 = 2s$ 时，

同理质点位置为 $(4m, 8m)$

$V_x = 2\,m/s$，$V_y = 16\,m/s$

$V = 2\sqrt{65}\,m/s$

方向与x轴夹角为α 且 $\tan\alpha = \frac{1}{8}$

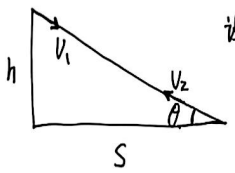
$a_x = 0\,m/s^2$ $a_y = 8\,m/s^2$

$a = 8\,m/s^2$

方向沿y轴正方向

$\begin{cases} \vec{r}_2 = 4\vec{i} + 8\vec{j} \\ \vec{V}_2 = 2\vec{i} + 16\vec{j} \\ \vec{a}_2 = 8\vec{j} \end{cases}$

1.12



设 $\tan\theta = \frac{h}{S}$

由题知 $V_1 = V_2 = V$

$V_{1x} = V_1\cos\theta$ ①

$V_{1y} = V_1\sin\theta$ ②

$V_{2x} = -V_2\cos\theta$ ③

$V_{2y} = -V_2\sin\theta$ ④

假设炮弹会相碰则

$(V_{1x} - V_{2x})t = S$ ⑤

$h - (V_{1y}t + \frac{1}{2}gt^2) = -(V_{2y}t + \frac{1}{2}gt^2)$ ⑥ 由 $\tan\theta = \frac{h}{S}$ 知

解 $(V_1 + V_2)\cos\theta \cdot t = S$ 显然恒成立

$(V_1 + V_2)\sin\theta \cdot t = h$

故两枚炮弹一定会在空中相碰

由⑤可知 $t = \frac{S}{2V\cos\theta}$

又 $h \geq V_{1y}t + \frac{1}{2}gt^2$

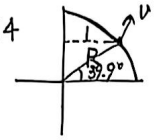
解 $V \geq 45.63\,m/s$

扫描全能王 创建

作业纸

班级：　　姓名：　　学号：　　科目：　　第　页

1.14



地球自转角速度 $\omega = \frac{2\pi}{T} = \frac{2\pi}{24\times3600s} = 7.27\times10^{-5}\ rad/s$

天安门随地球自转的速度

$v = \omega\cdot l$

$l = R\cdot\cos\alpha$

$\alpha = 39.9°$，$R = 6371.393\times10^{3}m$

解 $v = 355.35\ m/s$

易知 天安门的加速度只有向心加速度

$ma = m\frac{v^2}{l}$

解 $a = 0.026\ m/s^2$

1.18 (1) 由光盘致密可知

$$T = \int_{R_1}^{R_2}\frac{2\pi rN\cdot dr}{v} = \frac{\pi N(R_2^2-R_1^2)}{v} = 4163.64s$$

(2) $\omega = \frac{v}{r}$　　解 $\omega = 26\ rad/s$

$\dot{\omega} = \frac{v}{\dot{r}}\cdot(-\frac{1}{r^2}) = -\frac{v}{r^2}\cdot\frac{1}{\dot{r}}$

$\int_{R_1}^{r} 2\pi xN\,dx = vt$

两边同时对 t 求导得 $2\pi Nr\cdot\dot{r} = v$

解 $\dot{\omega} = -\frac{v^2}{2\pi Nr^3}$

代入数据有 $\dot{\omega} = -3.312\times10^{-3}\ rad/s^2$

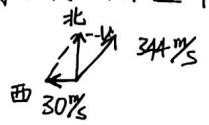
即角速度为 $26\ rad/s$

角加速度大小为 $3.312\times10^{-3}\ rad/s^2$，方向与角速度相反

1.20 声音向东传播的速率是 $374\ m/s$

向西传播的速率是 $314\ m/s$

设向北传播的速率是 v



$v = \sqrt{344^2-30^2}\ m/s$

$= 342.69\ m/s$

扫描全能王 创建

1.21 0～0.5s时，电梯下落的距离 $h_0=\frac{1}{2}at_0^2$ ①

0.5s时，电梯的下落速度 $v_0=at_0$ ②

设小球落到底板上所需时间为t则

$v_0t+\frac{1}{2}gt^2=v_0t+\frac{1}{2}at^2+\Delta h$ ③

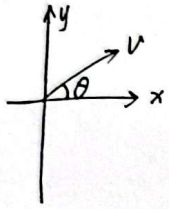
解 $t=0.59s$

小球对地面下落的距离

$h=v_0t+\frac{1}{2}gt^2$ ④

解：$h=2.06m$

加题1：



$V_x=V\cos\theta$ ①

$V_y=V\sin\theta$ ②

$x=V_xt$ ③

$y=V_yt-\frac{1}{2}gt^2$ ④

解：$\begin{cases} x=V\cos\theta\, t \\ y=V\sin\theta\, t-\frac{1}{2}gt^2 \end{cases}$

$\dot{x}=v\cos\theta$ ⑤ $\ddot{x}=0$ ⑦

$\dot{y}=v\sin\theta-gt$ ⑥ $\ddot{y}=-g$ ⑧

曲率半径 $R=\dfrac{[(v\cos\theta)^2+(v\sin\theta-gt)^2]^{\frac{3}{2}}}{|-gv\cos\theta|}$ ⑨

解：$R=\dfrac{(v^2-2gv\sin\theta\, t+g^2t^2)^{\frac{3}{2}}}{gv\cos\theta}$

初始点：$t=0$ 代入R可知 $R=\frac{v^2}{g\cos\theta}$

最高点：$t=\frac{v\sin\theta}{g}$ 代入R可知 $R=\frac{v^2\cos^2\theta}{g}$

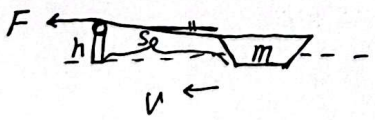
加题2：

绳子缩短的长度 $\Delta l=\sqrt{h^2+s_0^2}-\sqrt{h^2+s^2}$ ①

人拉力做的功 $W=F\cdot\Delta l$ ②

由动能定理有

$W=\frac{1}{2}mv^2$ ③



$V_{绳}=V\cdot\cos\theta$ ④

$\tan\theta=\frac{h}{s}$ ⑤

解 $V_{绳}=\sqrt{\dfrac{2F(\sqrt{h^2+s_0^2}-\sqrt{h^2+s^2})s^2}{m(h^2+s^2)}}$

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